

Plasmid expression of lasB from PA14 in pBBR1MCS2

Primer design • PCR / Assembly protocol • 2026-05-11

1 Konstrukt-Überblick

Zielgen	lasB in PA14 (1497 nt, 498 aa)
Strategie	Plasmid expression (coding sequence 1524 nt)
Vektor	pBBR1MCS2 (5148 bp), HindIII-linearisiert
Methode	In-Fusion 2-Fragment (Vektor + Insert)
Counterselektion	Km-Selektion (pBBR1MCS-2)
Finales Plasmid	pBBR1MCS2-lasB-His6-C_PA14 • 6690 bp

2 Primer

Primer	Sequenz 5'→ 3' (Tail fett, Body normal)	nt	Tm _{body}	GC
P1_lasB_INSERT_Fwd	CGGTATCGATAAGCTAGGAGGACTTGTTTCATGAAGAAGGTTTCTACGCTTGACC	54	59.5 °C	44 %
P2_lasB_INSERT_Rev	ATTCGATATCAAGCTTTAGTGGTGATGATGATGG	34	51.2 °C	42 %

3 PCR-Bedingungen (B7 High Fidelity)

Tm _{body} Mittel	55.4 °C (Spread 8.29 °C)
Annealing Ta	55 °C (= Tm _{body} ; Gradient ± 5 °C empfohlen falls 1. Versuch fehlschlägt)
Elongation	60 s/kb • Insert 1568 bp → 94 s

4 Konstrukt-Zusammenfassung

Insert (PCR-Amplikon) 1568 bp • Final 6690 bp

Off-Target Scan: PASS • HindIII-Sites in finalem Plasmid: 1 (erwartet 1)